

Statistics 6910, Section 003, Spring 2013, Quiz 1 (25 Points)

Friday, January 18, 2013

Your Name: _____

Question 1: Complete the Text (8 Points)

What is entered at the prompt in R is called an expression. We say that R _____ the expression (i.e., R performs a computation). To know what computation to perform, R breaks down an expression into small parts, called _____. These are elementary building blocks that can't be broken down any further. The inputs to function-style expressions are called _____ and the output is called _____.

Choose from the following options: computes, determines, evaluates, arguments, code-pieces, defaults, number, result, return value, snippets, tokens, values.

Question 2: Normal Curve (7 Points)

Suppose we want to generate 10 random values from a normal curve with mean $\mu = 5$ and standard deviation $\sigma = 8$. Which of the following commands is suitable for this task? **Circle the correct one(s).** We start with 7 points. For each missing correct answer, you will lose points — and for each additional incorrect answer, you will also lose points. Not marking anything at all will result in 0 points.

`rnorm(mean = 5, var = 8^2, n = 10)`

`rnorm(5, 8, 10)`

`rnorm(10, 5, 8)`

`rnorm(5, 64, 10)`

`rnorm(10, 5, 64)`

`rnorm(n = 10, 5, var = 8^2)`

`rnorm(n = 10, 5, var = 64)`

`rnorm(5, 5, 8, n = 10)`

`rnorm(5, 5, 8, 10)`

`rnorm(10, 5, sd = 8)`

Question 3: Formatting (10 Points)

Assume that in the following R code, each of the 3 consecutive lines (starting with the assignment to a variable) will be marked simultaneously in RStudio and we then click on “Run”. What will be the content of `x1`, `...`, `x5`? Just write down the value to the right of the variable. If an error is going to occur and nothing gets assigned to a variable, then state *undefined*.

```
x1 <- 1 +2 +3  
+4 +5 +6  
x1
```

```
x2 <- 1 + 2 + 3 +  
4 + 5 + 6  
x2
```

```
x3 <- 1 +2 +3  
+4 + 5 + 6)  
x3
```

```
x4 <- (1 +2 +3  
+4 + 5 + 6  
x4
```

```
x5 <- (1 +2 +3  
+4 + 5 + 6)  
x5
```

Extra Credit (2 Points): What happens in this case? Look carefully!

```
x2 < -1 + 2 + 3 +  
4 + 5 + 6
```